Multiplying by One Digit

Break the big number into tens and ones. Multiply each part. Add.

23 × 4: break 23 into 20 and 3

	20	3
4	80	12

$4 \times 20 = 80$

$4 \times 3 = 12$

$80 + 12 = 92$

Three digits work the same way: 146 × 3

	100	40	6
3	300	120	18

$300 + 120 + 18 = 438$

Hundreds, tens and ones.

No picture? Write the parts in a list

$5 \times 30 = 150$

$5 \times 7 = 35$

$150 + 35 = 185$

37 × 5 = 185

Tens times 5, ones times 5, then add.

Your turn!

	30	4
2	60	8

68

For teachers

Goal: multiply a two- or three-digit number by one digit with the area model, then partial products (4.NBT.5). It is packet 38's break-it-apart area with packet 37's tens. Say “Tens times 4, ones times 4, then add.”

Multiplying by One Digit: Practice

Break it apart. Multiply each part. Add the parts.

Part 1 The area model The parts are on top. Multiply each part.

1) **26 × 3 = 78**

	20	6
3	60	18

60 + 18 = 78

2) **42 × 5 = 210**

	40	2
5	200	10

200 + 10 = 210

3) **17 × 6 = 102**

	10	7
6	60	42

60 + 42 = 102

Multiplying by One Digit: Practice (continued)

Part 1 The area model (continued)

4) **35 × 4 = 140**



120 + 20 = 140

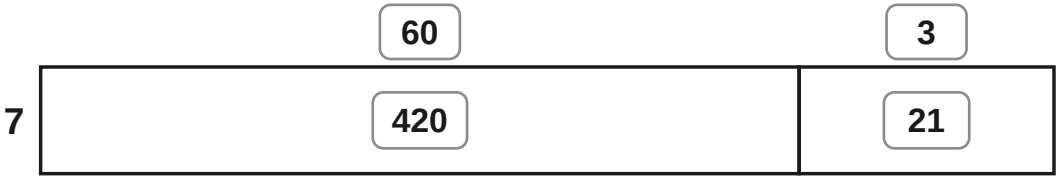
Part 2 Split it yourself Write the tens and ones on top first.

5) **48 × 2 = 96**



80 + 16 = 96

6) **63 × 7 = 441**

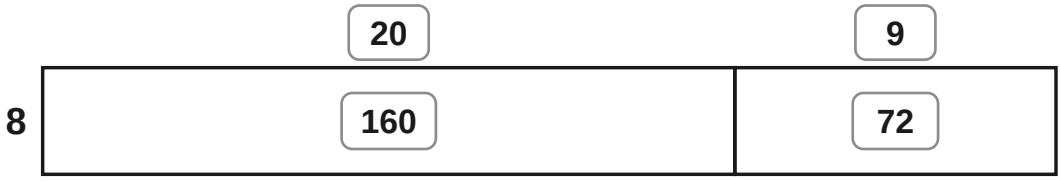


420 + 21 = 441

Multiplying by One Digit: Practice (continued)

Part 2 Split it yourself (continued)

7) **29 × 8 = 232**



160 + 72 = 232

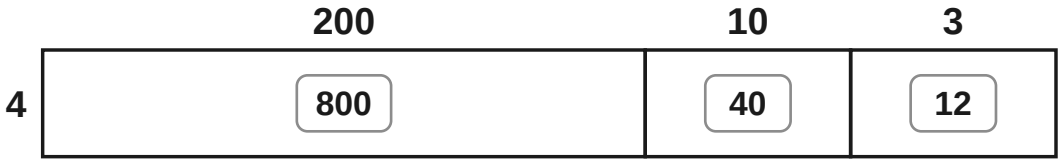
8) **54 × 6 = 324**



300 + 24 = 324

Part 3 Three digits Hundreds, tens and ones.

9) **213 × 4 = 852**



800 + 40 + 12 = 852

Multiplying by One Digit: Practice (continued)

Part 3 Three digits (continued)

10) **152 × 6 = 912**

	100	50	2
6	600	300	12

600 + 300 + 12 = 912

11) **324 × 3 = 972**

	300	20	4
3	900	60	12

900 + 60 + 12 = 972

12) **241 × 5 = 1,205**

	200	40	1
5	1,000	200	5

1,000 + 200 + 5 = 1,205

Multiplying by One Digit: Practice (continued)

Part 4 Write the parts No picture this time.

13) **36 × 4**

4 × 30 = 120

4 × 6 = 24

120 + 24 = 144

14) **71 × 3**

3 × 70 = 210

3 × 1 = 3

210 + 3 = 213

15) **58 × 6**

6 × 50 = 300

6 × 8 = 48

300 + 48 = 348

16) **124 × 7**

7 × 100 = 700

7 × 20 = 140

7 × 4 = 28

700 + 140 + 28 = 868

Part 5 Two digits No help this time.

17) 32 × 4 = 128

18) 57 × 3 = 171

19) 46 × 5 = 230

20) 83 × 2 = 166

21) 19 × 7 = 133

22) 64 × 3 = 192

Multiplying by One Digit: Practice (continued)

Part 5 Two digits (continued)

23) 38 × 6 = 228

24) 75 × 4 = 300

25) 91 × 8 = 728

26) 27 × 9 = 243

27) 44 × 7 = 308

28) 66 × 5 = 330

Part 6 Three digits No help this time.

29) 214 × 3 = 642

30) 132 × 4 = 528

31) 305 × 6 = 1,830

32) 421 × 5 = 2,105

33) 163 × 2 = 326

34) 247 × 3 = 741

Multiplying by One Digit: Practice (continued)

Part 6 Three digits (continued)

35) 512 × 7 = 3,584

36) 138 × 4 = 552

Part 7 Multiply

37) 55 × 3 = 165

38) 94 × 9 = 846

39) 69 × 9 = 621

40) 81 × 9 = 729

41) 88 × 3 = 264

42) 38 × 4 = 152

43) 49 × 3 = 147

44) 63 × 4 = 252

45) 77 × 8 = 616

Multiplying by One Digit: Practice (continued)

Part 7 Multiply (continued)

46) $22 \times 8 = 176$	47) $86 \times 8 = 688$	48) $54 \times 8 = 432$
49) $98 \times 5 = 490$	50) $53 \times 4 = 212$	51) $95 \times 8 = 760$
52) $17 \times 7 = 119$	53) $33 \times 5 = 165$	54) $73 \times 8 = 584$
55) $88 \times 6 = 528$	56) $85 \times 9 = 765$	57) $18 \times 8 = 144$

Multiplying by One Digit: Practice (continued)

Part 7 Multiply (continued)

58) 29 × 9 = 261

59) 42 × 5 = 210

60) 29 × 4 = 116

Part 8 Bigger numbers

61) 582 × 5 = 2,910

62) 436 × 7 = 3,052

63) 852 × 8 = 6,816

64) 209 × 9 = 1,881

65) 530 × 4 = 2,120

66) 637 × 8 = 5,096

67) 184 × 4 = 736

68) 540 × 8 = 4,320

Multiplying by One Digit: Practice (continued)

Part 8 Bigger numbers (continued)

69) 544 × 3 = 1,632

70) 483 × 5 = 2,415

71) 359 × 4 = 1,436

72) 504 × 3 = 1,512

73) 393 × 3 = 1,179

74) 181 × 7 = 1,267

75) 824 × 3 = 2,472

76) 868 × 4 = 3,472

77) 289 × 6 = 1,734

78) 576 × 7 = 4,032

Multiplying by One Digit: Practice (continued)

Part 9 Remember? Problems from earlier packets.

79) **3,846 to the nearest hundred =** 3,800

80) **2,457 + 1,386 =** 3,843

81) **in 7,062 the 7 is worth** 7,000

82) **4 × 20 =** 80

83) **6 L + 9 L =** 15 **L**

84) **5,000 − 1,246 =** 3,754

Part 10 Word problems Draw the area model if it helps.

85) A box holds 24 crayons. How many crayons are in 6 boxes?

Number sentence 24 × 6 = 144

Answer 144 crayons

Draw or show your work

86) A bus has 48 seats. How many seats are on 4 buses?

Number sentence 48 × 4 = 192

Answer 192 seats

Draw or show your work

87) Tickets cost $35 each. How much do 4 tickets cost?

Number sentence 35 × 4 = 140

Answer $140

Draw or show your work

Multiplying by One Digit: Practice (continued)

Part 10 Word problems (continued)

88) A farmer plants 7 rows of 126 corn plants. How many plants is that?

Number sentence **126 × 7 = 882**

Answer **882** plants

Draw or show your work

89) Challenge: A school has 5 classes of 28 kids and 4 classes of 31 kids. How many kids in all?

Number sentence **28 × 5 = 140, 31 × 4 = 124, 140 + 124 = 264**

Answer **264** kids

Draw or show your work

Check-In: Multiplying by One Digit

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 The area model

1) **41 × 4 = 164**



160 + 4 = 164

Part 2 Write the parts

2) **67 × 4**

4 × 60 = 240

4 × 7 = 28

240 + 28 = 268

3) **89 × 5**

5 × 80 = 400

5 × 9 = 45

400 + 45 = 445

Part 3 Multiply

4) **29 × 6 = 174**

5) **45 × 5 = 225**

6) **96 × 9 = 864**

Part 4 Three digits

7) **431 × 4 = 1,724**

8) **199 × 7 = 1,393**